

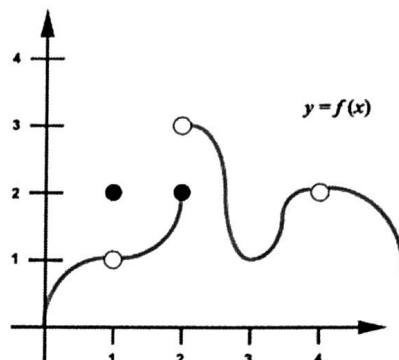
1. Evaluate the following limits based on the picture below or state that it does not exist. [1 each]

$\lim_{x \rightarrow 4} f(x) = \underline{2}$ $f(4) = \underline{\text{DNE}}$

$\lim_{x \rightarrow 1} f(x) = \underline{1}$ $\lim_{x \rightarrow 2^-} f(x) = \underline{2}$

$\lim_{x \rightarrow 2} f(x) = \underline{\text{DNE}}$ $\lim_{x \rightarrow 0} f(\cos(x) + 1) = \underline{2}$

$f(1+1) = f(2)$



$\frac{x+2}{\sqrt{x}}$

$16 \rightarrow 40$
4 4 4 4

4cm = 10 ft

b) Briefly explain, using the Formal Definition of continuity as your guide, why $f(x)$ is not continuous at $x = 2$. Use words and proper math notation. [1]

because $\lim_{x \rightarrow 2} f(x) = \text{DNE}$, $f(x)$ cannot be continuous.

2. Evaluate the following 3 limits algebraically or state that they do not exist. No work required (though you may do as much work as you'd like!) [2 each]

a) $\lim_{x \rightarrow 0^+} \frac{x+2}{\sqrt{x}} = \underline{\text{DNE}}$
 $\frac{x\sqrt{x} + 2\sqrt{x}}{x} = \frac{x^{\frac{3}{2}} + 2\sqrt{x}}{x}$ top heavy?

b) $f(x) = \begin{cases} x^2 + 3x & x < 2 \\ x^3 + 2 & x \geq 2 \end{cases}$ $\lim_{x \rightarrow 2} f(x) = \underline{10}$

$x^2 + 3x = x^3 + 2$

$x^3 - x^2 - 3x + 2 = 0$

$8 - 4 - 6 + 2 = 0$

$\frac{1}{x} - \frac{1}{4} = \frac{4-x}{4x}$

$4x \times \frac{1}{3x} = \frac{4x}{3x} = 1$

c) $\lim_{x \rightarrow \infty} \frac{\sqrt{x+5}}{x-4} = \underline{0}$
bottom heavy?

3. Evaluate the following showing your algebraic justifications. [3 each]

a) $\lim_{x \rightarrow 4} \frac{2x^3 - 5x^2 - 4x + 3}{x-3}$

$\begin{array}{r|rrrr} 3 & 2 & -5 & -4 & 3 \\ & & 6 & 3 & -3 \\ \hline & 2 & 1 & -1 & 0 \end{array}$

$= \lim_{x \rightarrow 4} (2x^2 + x - 1)$

$= 2(4)^2 + 4 - 1$

$= 2(16) + 3$

$= 32 + 3 = 35$

35 ✓

b) $\lim_{x \rightarrow 4} \frac{\frac{1}{x+2} - \frac{2}{3x}}{\frac{1}{x} - \frac{1}{4}} = \lim_{x \rightarrow 4} \left(\frac{3x - 2(x+2)}{(x+2)(3x)} \times \frac{4x}{4-x} \right)$

$= \lim_{x \rightarrow 4} \left(\frac{3x - 2x - 4}{(x+2)(3x)} \times \frac{4x}{4-x} \right) = \lim_{x \rightarrow 4} \left(\frac{(x-4)(4x)}{(x+2) \cdot 1 \cdot (x-4)(3x)} \right)$

$= \lim_{x \rightarrow 4} \left(\frac{4x}{-(x+2)(3x)} \right) = \lim_{x \rightarrow 4} \left(\frac{1}{-(x+2)} \right) = \frac{1}{-(4+2)} = -\frac{1}{6}$

$-\frac{1}{6} \cdot \frac{4}{3} = -\frac{2}{9}$

$2(64) - 5(16) - 4(4) + 3$

$= 128 - 80 - 16 + 3$

-1

$$\left(\frac{3\pi}{4} - \epsilon\right)\left(\frac{1}{3}\right) = \frac{3\pi}{12} - \frac{\epsilon}{3} = \frac{\pi}{4} - \frac{\epsilon}{3}$$

4. To the right is a portion of the graph of

$$f(x) = 3 \arctan(x) + \frac{\pi}{4}$$

a) State $\lim_{x \rightarrow 1} f(x) = \pi$ [1]

b) Perform a delta epsilon proof to prove your statement above. Be sure to justify any assumptions you make along the way and end with a conclusion statement. [4]

will find smallest δ value on the left

$$\pi - \epsilon = 3 \arctan(x) + \frac{\pi}{4}$$

$$3 \arctan(x) = \frac{3\pi}{4} - \epsilon$$

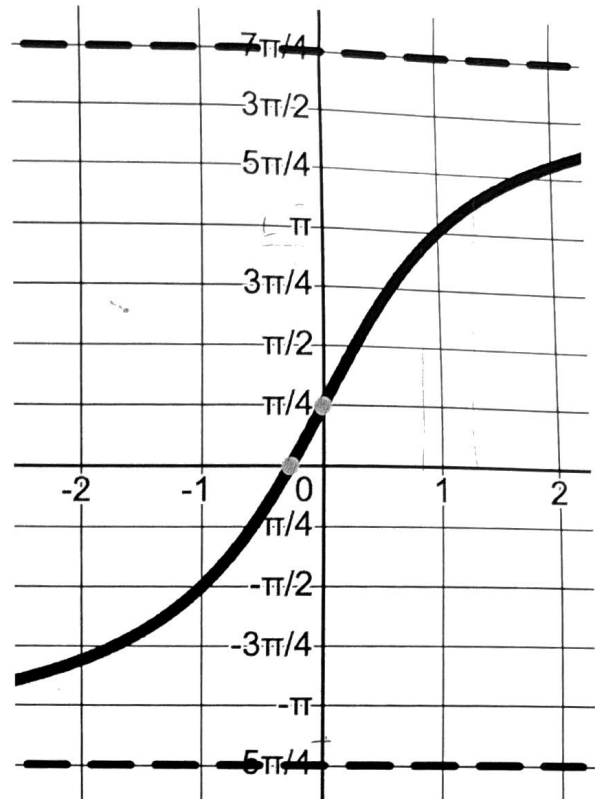
$$\arctan(x) = \left(\frac{3\pi}{4} - \epsilon\right)\left(\frac{1}{3}\right) = \frac{\pi}{4} - \frac{\epsilon}{3}$$

$$x = \tan\left(\frac{\pi}{4} - \frac{\epsilon}{3}\right) \quad \delta = 1 - \tan\left(\frac{\pi}{4} - \frac{\epsilon}{3}\right)$$

given $\epsilon > 0$, when x is within $1 - \tan\left(\frac{\pi}{4} - \frac{\epsilon}{3}\right)$ of 1, $f(x)$ is within ϵ of π . Thus,

$$\lim_{x \rightarrow 1} (3 \arctan(x) + \frac{\pi}{4}) = \pi.$$

$$\frac{1}{100} \times \frac{1}{3} = \frac{1}{300}$$



c) The $\lim_{x \rightarrow -\infty} f(x) = \frac{-5\pi}{4}$. Use $\epsilon = .01$ to convince me that this is true by finding a corresponding δ and writing a conclusion statement. [3]

$$\frac{-5\pi}{4} + 0.01 = 3 \arctan(x) + \frac{\pi}{4} \quad \arctan(x) = -\frac{\pi}{2} + \frac{1}{300}$$

$$3 \arctan(x) = \frac{-3\pi}{2} + 0.01 \quad x = \tan\left(-\frac{\pi}{2} + \frac{1}{300}\right)$$

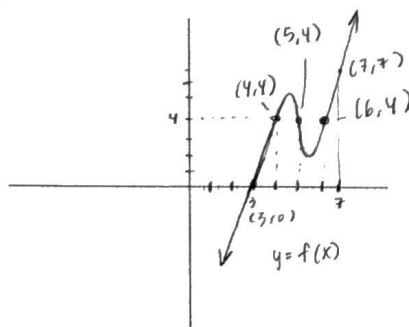
$$\arctan(x) = \left(\frac{-3\pi}{2} + 0.01\right)\left(\frac{1}{3}\right)$$

given $\epsilon = 0.01$, when x is less than $\tan\left(-\frac{\pi}{2} + \frac{1}{300}\right)$, then $f(x)$ is within 0.01 of $-\frac{5\pi}{4}$. Thus,

$$\lim_{x \rightarrow -\infty} f(x) = -\frac{5\pi}{4}$$

5. In this problem you are going to draw 2 sketches illustrating your understanding of the Intermediate Value Theorem for a function $f(x)$ over the x interval $[3, 7]$. [2pts each]

a) Sketch $f(x)$ such that it is continuous AND there are three possible "c" values over the x interval $[3, 7]$ for $y = 4$. Include labels on the axes for all relevant points.



b) Sketch $f(x)$ so that there are 2 x-intercepts between $x = 3$ and $x = 7$, even though the IVT does not guarantee any x-intercepts over that x interval. Again, label your axes.

